



Digital Electronics

Logic Gates: using DTL and modified DTL

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DTL Logic

- The basic circuit in the DTL digital logic family is the NAND gate.
- Each input is associated with one diode.
- The diodes and the 5-k Ω resistor form an AND gate.
- The transistor serves as a current amplifier while inverting the digital signal.
- The two voltage levels are 0.2 V for the low level and between 4 and 5 V for the high level.
- The power dissipation of a DTL gate is about 12 mW.
- The propagation delay averages 30 ns.
- The noise margin is about 1 V and a fan-out as high as 8 is possible.



DTL Logic - Disadvantages

- Relatively lower speed.
- Propagation is higher than RTL.



NAND Gate Using DTL Logic

- If any input of the gate is low at 0.2 V, the corresponding input diode conducts current through V_{CC} and the 5-k Ω resistor into the input node.
- The voltage at point P is equal to the input voltage of 0.2 V plus a diode drop of 0.7 V, for a total of 0.9 V.

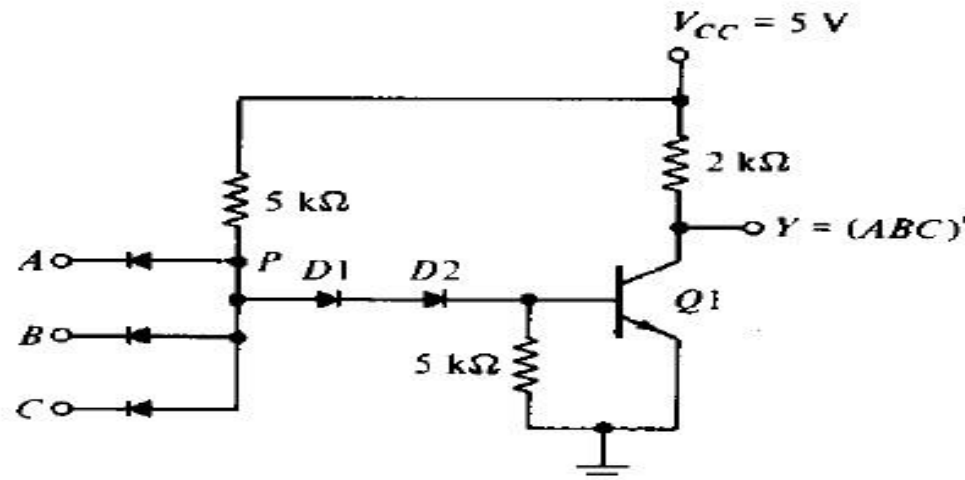


FIGURE
DTL basic NAND gate



NAND Gate Using DTL Logic

- In order for the transistor to start conducting, the voltage at point P must overcome a potential of one V_{BE} drop in Q_1 plus two diode drops across D_1 and D_2 , or $3 \times 0.6 = 1.8$ V.
- Since the voltage at P is maintained at 0.9 V by the input conducting diode, the transistor is cut off and the output voltage is high at 5 V.
- If all inputs of the gate are high, the transistor is driven into the saturation region.
- The voltage at P now is equal to V_{BE} plus the two diode drops across D_1 and D_2 , or $0.7 \times 3 = 2.1$ V.
- Since all inputs are high at 5 V and $V_P = 2.1$ V, the input diodes are reverse biased and off.
- The base current is equal to the difference of currents flowing in the two 5-k Ω resistors and is sufficient to drive the transistor into saturation.
- With the transistor saturated, the output drops to V_{CE} of 0.2 V, which is the low level for the gate.



NAND Gate Using DTL Logic

Operation

**For Q1 to conduct, $V_p > V_{BE}(Q1) + (2 \times 0.7)$
 $= 0.7 \text{ V} + 1.4 \text{ V} = 2.1 \text{ V}$

Input	Transistor & Diode Status	Output
Any Input = Low (0.2V)	Corresponding input diode forward biased. $V_p = (0.2 + 0.7) \text{ V}$	As $V_p < 2.1 \text{ V}$, Q1 is OFF. Cutoff region. $Y = V_{CC} = 5 \text{ V}$ (High)
All input = High (5 V)	All input diode reverse biased. D1 & D2 forward biased. Q1 ON. Saturation region. $V_p = V_{BE}(Q1) + V_{D1} + V_{D2}$ $= 0.7 + 0.7 + 0.7 = 2.1 \text{ V}$	Q1 is ON. Saturation region. $Y = V_{CE} = 0.2 \text{ V}$ (Low)

Circuit

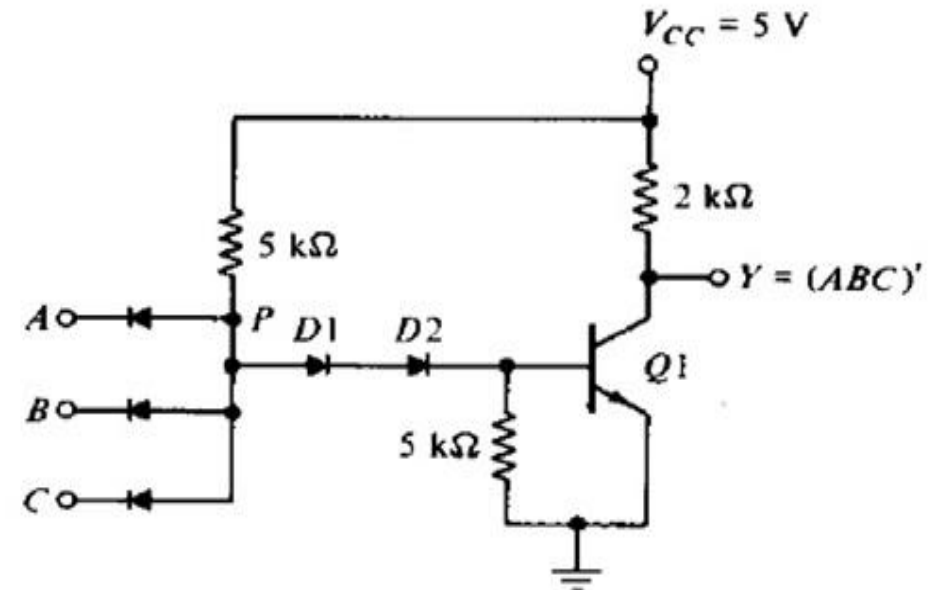


FIGURE
DTL basic NAND gate



NAND Gate Using modified DTL Logic

- The fan-out of the DTL gate is limited by the maximum current that can flow in the collector of the saturated transistor.
- The fan-out of a DTL gate may be increased by replacing one of the diodes in the base circuit with a transistor.

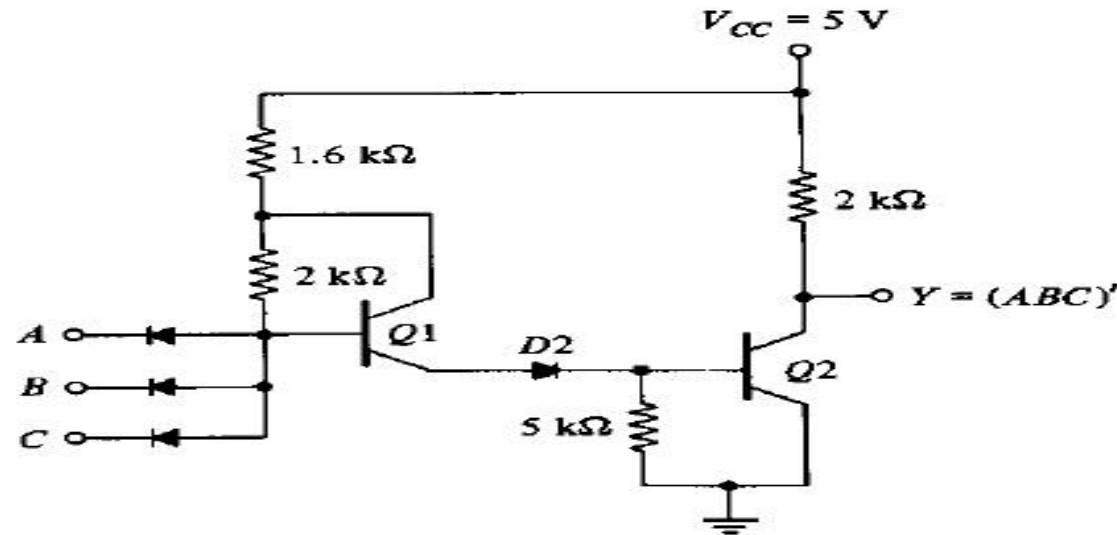


FIGURE
Modified DTL gate



NAND Gate Using modified DTL Logic

- Transistor $Q1$ is maintained in the active region when output transistor $Q2$ is saturated.
- As a consequence, the modified circuit can supply a larger amount of base current to the output transistor.
- The output transistor can now draw a larger amount of collector current before it goes out of saturation.
- Part of the collector current comes from the conducting diodes in the loading gates when $Q2$ is saturated.
- Thus, an increase in allowable collector saturated current allows more loads to be connected to the output, which increases the fan-out capability of the gate.



NAND Gate Using HTL Logic

- Due to presence of electrical motor's on-off control circuits, High voltage switches etc. are used in industrial environment, Thereby noise level is very high.
- So DTL redesigned with a power supply of 5V and the D2 is replaced by a zener diode with a breakdown voltage of 6.9 V.
- So HTL posses a high threshold of noise immunity.
- To conduct Q2, the emitter of Q1 must rise to a potential,
$$V_{BE}(Q2) + V_Z = 0.7 + 6.9 = 7.6 \text{ V}$$
- So, Only if the noise signal is greater than 7.6 V , then it will be able to change the state. Higher noise immunity.

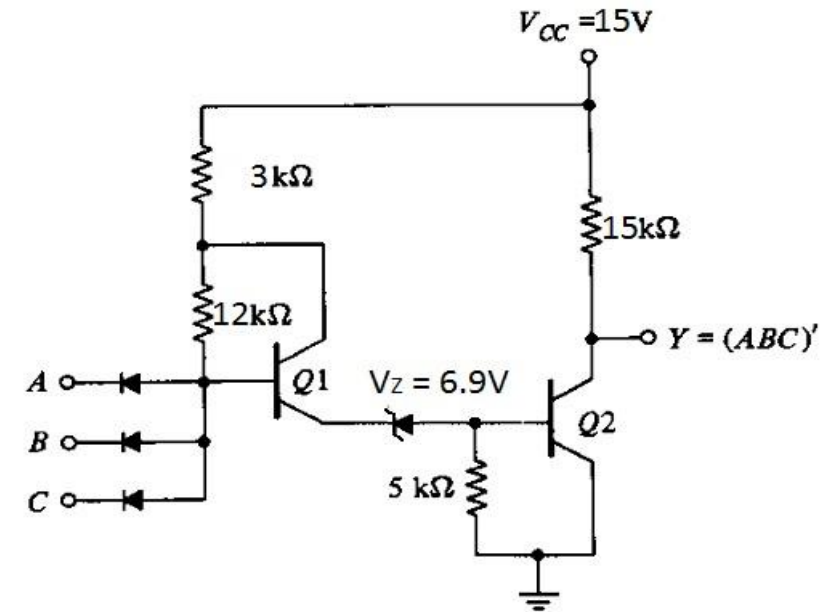


Figure: 3 Input HTL NAND



References

1. Thomas L. Floyd, “Digital Fundamentals” 8th edition, Prentice Hall – Pearson Education.
2. M. Morris Mano, “Digital Logic & Computer Design” Prentice Hall